

Assignment 9

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1 Dataset and Experimental Setup

The experiment used OpenFWI velocity model data stored in:

`model-20260504T122211Z-3-001/model`

The notebook loaded all 60 files, from `model11.npy` to `model160.npy`. Each file contains 500 velocity models with shape $(500, 1, 70, 70)$, so the full run used 30000 velocity models in total.

The data were split into 27000 training samples and 3000 validation samples. The original velocity values ranged from 1500.0 to 4500.0, and all models were normalized to $[0, 1]$ before training.

The experiment was run on CUDA. The default hyperparameter setting was:

- latent dimension: 64,
- batch size: 64,
- VAE training epochs: 10,
- GAN training epochs: 10.

2 Models Compared

Four models were trained and evaluated:

- **MLP-VAE**
- **CNN-VAE**
- **MLP-GAN**

- **CNN-GAN**

The MLP models treat each velocity model as a flattened vector, while the CNN models preserve the two-dimensional spatial layout. Because OpenFWI velocity models are structured images with clear local continuity, the CNN variants are expected to capture spatial patterns more effectively.

3 Training Results

3.1 VAE Results

For the VAE models, the reported training metric was validation mean squared error (MSE).

The MLP-VAE validation MSE decreased from 0.009 at epoch 1 to 0.004 at epoch 10. The CNN-VAE performed better, improving from 0.006 at epoch 1 to 0.001 at epoch 10.

These results show that the CNN-VAE reconstructed the OpenFWI velocity models more accurately than the MLP-VAE by the end of training.

3.2 GAN Results

For the GAN models, discriminator loss and generator loss were recorded.

The MLP-GAN finished with discriminator loss 0.668 and generator loss 0.857. The CNN-GAN finished with discriminator loss 0.179 and generator loss 5.444.

Although GAN losses are not directly comparable to VAE reconstruction errors, they provide useful information about adversarial training behavior and are part of the saved notebook output.

4 Final Quantitative Comparison

The final comparison used **generation_score**, which combines value statistics, histogram difference, and spatial-gradient difference between generated and real velocity models. A lower score indicates that the generated samples are closer to the real OpenFWI data distribution.

Model	Family	Generation Score	Stats Error	Hist Error	Grad Error	Val MSE	D Loss	G Loss
CNN-VAE	VAE	0.029	0.011	0.006	0.011	0.001	–	–
CNN-GAN	GAN	0.032	0.013	0.008	0.011	–	0.179	5.444
MLP-GAN	GAN	0.049	0.006	0.005	0.038	–	0.668	0.857
MLP-VAE	VAE	0.109	0.078	0.016	0.015	0.004	–	–

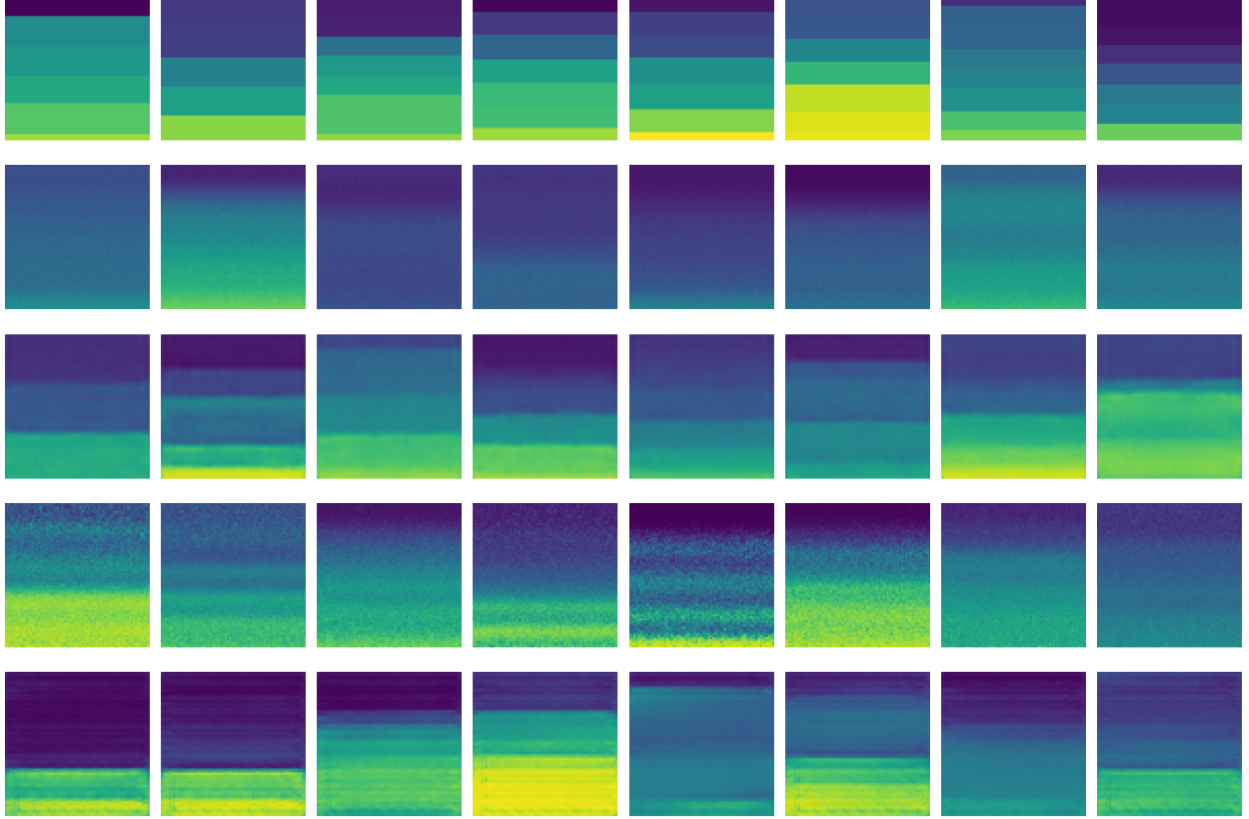


Figure 1: Qualitative comparison of real and generated OpenFWI velocity models. From top to bottom, the rows show real samples, MLP-VAE samples, CNN-VAE samples, MLP-GAN samples, and CNN-GAN samples. Each column corresponds to one sample position across the compared sets. The visual patterns are consistent with the updated quantitative ranking: CNN-based models preserve layered spatial structure better than MLP-based models, and CNN-VAE provides the best overall match in this run.

Among all four models, **CNN-VAE** achieved the lowest generation score, 0.029 , which makes it the best generator in this run according to the output. Figure 1 provides the corresponding qualitative comparison.

5 Conclusion

The results support two main observations.

First, the CNN-based models performed better than the MLP-based models overall. This is reasonable because OpenFWI velocity models are two-dimensional spatial images, and convolutional layers are better suited to preserving local structure, boundaries, and continuity than flattened MLP inputs.

Second, the best final generator was **CNN-VAE**, not a GAN model. While the CNN-GAN remained competitive, the generation-score ranking shows that CNN-VAE produced samples that more closely matched the real velocity-model distribution in the run. The slightly lower GAN performance may also be related to the limited training schedule: with only 10 epochs, the GAN models may not have fully converged yet, so additional training could further improve their generation quality.

The main conclusions are as follows:

- CNN architectures are more effective than MLP architectures for OpenFWI velocity-model generation.
- CNN-VAE reconstructs the data more accurately than MLP-VAE.
- CNN-VAE achieves the best final generation quality, with the lowest generation score of 0.029 .
- Therefore, the optimal model for this experiment is **CNN-VAE**.

6 Code

https://github.com/WKKKKKKKKKKKKKK/ML_IN_GEO